Meridian Multimodal Research Platform

The finalized 34-step protocol: tasks, parameters, and their basis in the literature

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Abstract

Meridian records a multimodal dataset of emotional and physiological responses: contact sensors (ECG, finger oximeter, skin conductance, surface EMG), contactless sensors (two face cameras, thermal camera, Wi-Fi channel state information), and audio, all on one host clock. This document describes the **finalized shortened protocol (version 26)**: **34 steps in about 37 minutes**. Participants watch six emotion film clips, in one fixed order from the calmest to the most heart-rate-raising, and rate each one; perform one cognitive task between clips (Stroop or N-back, alternating between groups of six participants); take a word recognition test; read a fixed passage aloud; follow a controlled-breathing block that provides reference breathing rates for contactless sensing; raise their heart rate with two short self-paced exercises (stand-to-sit and step-up); recover for two minutes while heart-rate recovery is recorded; and finish with a guided relaxation. The controlled-breathing block follows the recovery. For each task it gives the exact parameters implemented in the software and the published work the design choice rests on.

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1 The protocol at a glance



Table 1 is the plan for participant 1, rendered by the software from the saved protocol version 26 (`lib/protocol-template.ts`, `renderPlan`). The clip order is the same for every participant; only which cognitive task is done changes (Section 2), so every participant has the same 34 steps. Untimed steps (questionnaires, ratings, tasks, word list, read-aloud) show the planning estimate set in the protocol builder; see Section 6 for the ones likely to run longer. Clip lengths differ between the clips assigned to different participants, so the total varies by a few minutes.

Table 1: The 34 finalized steps (participant 1). *Start* is the clock time from the start of the protocol; *Time* is the software's planning estimate. Clip steps include a 5 s tail after the clip.

#	Phase	Step	Start	Time	Role (key reference)
1	Getting settled	Sensor and synchronization check	0:00	0:30	Verify every stream; *operator advances*
2	Getting settled	Current mood (PANAS, start)	0:30	1:15	Pre-session mood [44]
3	Getting settled	Resting baseline, eyes open	1:45	2:00	Physiological reference [19, 41]; *BP reading*
4	Part 1	Emotion clip: surprise	3:45	1:35	Emotion induction [13]
5	Part 1	Self-report: surprise	5:20	0:40	Labels [7]
6	Part 1	Neutral washout	6:00	0:20	Limit carry-over [31]
7	Part 1	Emotion clip: disgust	6:20	1:45	Emotion induction
8	Part 1	Self-report: disgust	8:05	0:40	Labels
9	Part 1	Stroop colour-word task	8:45	1:30	Cognitive stress [40, 30]; *N-back (0:50) for half the cohort*
10	Part 1	Workload rating (NASA-TLX)	10:15	0:45	Perceived workload [14]
11	Part 1	Return-to-baseline gate	11:00	0:30	Limit carry-over [19]; *waits for HR and SCL*
12	Part 2	Emotion clip: happy	11:30	1:38	Emotion induction
13	Part 2	Self-report: happy	13:08	0:40	Labels
14	Part 2	Neutral washout	13:48	0:20	Limit carry-over [31]
15	Part 2	Emotion clip: sad	14:08	1:35	Emotion induction
16	Part 2	Self-report: sad	15:43	0:40	Labels
17	Part 2	Return-to-baseline gate	16:23	0:30	Limit carry-over [19]; *waits for HR and SCL*
18	Part 3	Emotion clip: fear	16:53	1:40	Emotion induction
19	Part 3	Self-report: fear	18:33	0:40	Labels
20	Part 3	Neutral washout	19:13	0:20	Limit carry-over [31]
21	Part 3	Emotion clip: anger	19:33	1:35	Emotion induction
22	Part 3	Self-report: anger	21:08	0:40	Labels
23	Part 3	Word list: study and recognise	21:48	0:30	Recognition memory [36]; *≈1:00 in practice*
24	Part 3	Read aloud	22:18	0:30	Read speech, fixed passage [9]; *≈1:00 in practice*

#	Phase	Step	Start	Time	Role (key reference)
25	Heart rate elevation	Stand-to-sit	22:48	1:00	Raise heart rate [5]; *skipped by screening*
26	Heart rate elevation	Step-up	23:48	2:00	Raise heart rate further [3]; *skipped by screening*
27	Recovery	Rest	25:48	2:00	Heart-rate recovery [10, 39]; *BP reading*
28	Breathing	Paced breathing: 6 per minute	27:48	1:00	Respiration ground truth [25]; *starts on Start press*
29	Breathing	Paced breathing: 20 per minute	28:48	0:42	Respiration ground truth; *starts on Start press*
30	Breathing	Box breathing	29:30	1:04	Guided slow breathing [2]; *starts on Start press*
31	Relaxation	Guided body scan	30:34	4:30	Low-arousal state [20]
32	Relaxation	Return to normal	35:04	0:20	Final resting reference; *BP reading*
33	Relaxation	Current mood (PANAS, end)	35:24	1:15	Post-session mood [44]
34	All done	Session complete	36:39	0:15	Confirm recordings saved
		Total		**36:54**	≈38 min in practice; ≈67 min per participant with setup and consent

Which cognitive task each participant does

Each participant does one cognitive task, at the end of Part 1, followed by a NASA-TLX rating. Parts 2 and 3 have no task. Which task it is alternates once every six participants:

Participants	Cognitive task (end of Part 1)
1–6, 13–18, 25–30	Stroop
7–12, 19–24, 31–36	N-back

So each task is done by half of the participants, and Stroop and N-back are compared *between* participants rather than within them. Earlier versions gave every participant both tasks (versions 19–23) or two of three, including a conversation (up to version 18).

2 Design choices that span the whole session

Emotion clip order: calmest first, heart-rate-raising last

Every participant sees the six clips in the same order, graded by how much each emotion typically raises heart rate, so that heart rate climbs across the three parts rather than jumping about:

Part	Clips (in order)	Typical heart-rate response	Mean HR change, d
1	surprise, disgust	brief orienting slow-down; no reliable change	–, 0.07
2	happy, sad	modest rise	0.33, 0.36
3	fear, anger	largest rise	0.49, 0.52

The ranking follows the largest meta-analysis of autonomic responses in laboratory emotion induction, 202 studies in all [35]: mean heart-rate increase from baseline was largest for anger and fear,

moderate for sadness and happiness, and absent for disgust. Surprise was too rarely studied to be meta-analysed; it is dominated by a brief orienting deceleration, and disgust at bodily or surgical content tends to slow the heart as well [23]. Sadness is the least predictable: deactivated, non-crying sadness lowers heart rate while tearful sadness raises it [23], so its place in the middle is a compromise.

A fixed order has a cost that counterbalancing (the Williams square used before version 19) avoided: each emotion always occupies the same position, so drift, fatigue, and habituation over the session cannot be separated from the emotion itself. The return-to-baseline gates between the parts limit carry-over but do not remove the position confound (Section 6).

Task assignment

The cognitive task alternates once every six participants: Stroop for participants 1–6, N-back for 7–12, and so on. The word list a participant studies is assigned by a Latin square over the participant's position in a six-participant cycle and the task row, so list difficulty cannot masquerade as an order effect.

Reproducibility

Stroop, N-back, and word-test trial lists are generated from a logged random seed, so every participant's exact trial sequence can be regenerated during analysis. Each session stores a copy of the protocol version and the clip files it was assigned, so later edits in the protocol builder never alter a recorded session.

Washout and return-to-baseline gates

Physiological responses to an emotional film outlast the film itself. The protocol therefore separates the two clips of a block with a 20 s neutral fixation (washout), and ends Parts 1 and 2 with a *return-to-baseline gate*. The gate needs heart rate within 5 bpm, and skin conductance within 10%, of the eyes-open resting baseline, held for 15 consecutive seconds, before the operator advances. Using a measured resting period as the reference, rather than a fixed pause, follows the cardiovascular baseline literature [19], and film-elicitation guidance recommends neutral periods between clips to limit carry-over [31].

Order of the closing block: exercise, recovery, then breathing

The closing block runs straight from the read-aloud into the two exercises, a two-minute recovery, the breathing conditions, and the body scan. Putting the exercise first means the two minutes straight after it are a still, seated recording: the window in which heart-rate recovery is conventionally measured [10]. The breathing block then starts about two minutes after the step-up, while heart rate and breathing drive may still be above rest; cardiac parasympathetic activity can take considerably longer than that to return [39]. The paced-breathing steps are therefore recorded during late recovery rather than at full rest, which should be kept in mind when they are used as the breathing-rate and slow-breathing HRV reference (Section 6). The body scan at the end extends the recovery into a deliberately low-arousal state.

3 The finalized tasks

3.1 Sensor and synchronization check (step 1)

Participant does	Sits still facing the screen.
Parameters	Operator-advanced, about 30 s.
Measures	Presence and plausibility of every required stream; trigger synchronization.
Why it is here	A preflight check before any data that matters is recorded. Not a research task in itself.

3.2 Mood and personality questionnaires (steps 2, 33; TIPI before the session)

Participant does	Answers PANAS (20 mood adjectives, 5-point scale) at the start and end of the session, and TIPI (10 items, 7-point scale) during consent, before recording starts.
Parameters	PANAS at steps 2 and 33, placed *before* the resting baseline so answering does not disturb it. TIPI is no longer a session step: it measures stable traits, so it is collected on paper or a tablet at consent, which takes it off the recorded clock. **[TODO: Decide where the TIPI answers are stored so they are joined to the session.]**
Measures	Positive and negative affect before and after the session; Big-Five personality traits.
Why it is here	PANAS is the standard brief measure of positive and negative affect [44]; the pre/post pair checks whether the session as a whole shifted mood. Because the end-of-session PANAS follows the body scan, it reflects the whole session rather than the emotion clips alone. TIPI measures the Big Five in ten items [12], enough to use personality as a covariate for individual differences in emotional reactivity without a long inventory. Only instruments usable without licence fees are included.

3.3 Resting baseline, eyes open (step 3)

Participant does	Looks at a fixation cross and breathes naturally for 120 s.
Parameters	Timer-advanced. A resting blood-pressure reading is taken after settling.
Measures	Resting heart rate, heart-rate variability, skin conductance level, breathing, and face and thermal appearance.
Why it is here	Every later response is expressed relative to the participant's own resting state. Heart-rate-variability standards and psychophysiology recommendations call for a controlled resting recording before any manipulation [41, 24]; a quiet, low-demand resting baseline is also the reference the gates return to [19].

3.4 Emotion film clips (steps 4, 7, 12, 15, 18, 21)

Participant does	Watches six short film clips full screen with sound, one per basic emotion: happiness, sadness, anger, fear, surprise, disgust.
Parameters	Durations: happy 93 s, sad 90 s, anger 90 s, fear 95 s, surprise 90 s, disgust 100 s. Four clips are trimmed segments of longer sources, cut by the research team to the emotionally central part: surprise 0:40–2:10, disgust 1:00–2:40, sad 0:58 to the end of the source (2:27.6), and anger 0:20–1:50 (anger and disgust come from the replacement clips). Each cut is re-encoded frame-accurately with a 0.5 s audio fade at both ends and no visual fade. All are H.264 1280×720, 30 fps, AAC 48 kHz. Onset is logged when the video actually starts playing. Clips are drawn from **[TODO: name of the validated film set, with citation]**.
Measures	Emotional responses in every physiological and behavioural channel, time-locked to the clip.
Why it is here	Film clips are the most widely used and best validated way to elicit discrete emotions in the laboratory: they are standardized, dynamic, and ethically acceptable [13, 31]. Validated databases supply normative ratings for each clip [32]. The six categories follow Ekman's basic emotions [11]. The same paradigm underlies the main affective-computing datasets, DEAP [22] and MAHNOB-HCI [37], which makes Meridian directly comparable to them.

3.5 Self-report after each clip (steps 5, 8, 13, 16, 19, 22)

Participant does	Rates valence, arousal, and dominance; chooses the main emotion felt (six basic emotions or neutral) and its intensity; says whether the clip was familiar.
Parameters	Self-Assessment Manikin on 9-point scales; intensity on 9 points; familiarity Yes/No/Not sure. About 40 s.
Measures	Per-clip ground-truth labels, both dimensional and categorical.
Why it is here	The Self-Assessment Manikin is a non-verbal, fast, and validated measure of valence, arousal, and dominance [7]. Collecting both dimensional and categorical labels, as DEAP and MAHNOB-HCI do [22, 37], lets the dataset support both kinds of models and checks whether each clip elicited its target emotion. Familiarity lets analysis exclude clips the participant had seen before.

3.6 Neutral washout (steps 6, 14, 20)

Participant does	Looks at a fixation cross and breathes normally.
Parameters	20 s, timer-advanced.
Measures	Decay of the response to the preceding clip.
Why it is here	Separates the two clips of a block to reduce carry-over between emotions [31].

3.7 Return-to-baseline gates (steps 11, 17)

Participant does	Sits quietly and breathes normally.
Parameters	At least 25 s; advances once heart rate and skin conductance are back near baseline (Section 2).
Measures	Whether physiology has recovered before the next block.
Why it is here	Prevents the response to one block from contaminating the next [19, 31].

3.8 Stroop colour-word task (step 9, half of the participants)

Participant does	Chooses the *ink colour* of a colour word, ignoring what the word says.
Parameters	30 trials, 15 congruent and 15 incongruent; no more than 3 trials in a row of the same kind; no immediate word repeats; six colours; 2 s response deadline; 400 ms fixation. Reaction time and accuracy per trial. The screen before the block tells the participant how many words there are and that each disappears after about two seconds, so the pace is known rather than discovered.
Measures	Selective attention and interference control; physiological response to a mild cognitive stressor.
Why it is here	The Stroop interference effect is one of the most replicated findings in cognitive psychology [40, 27]. Performing it under time pressure reliably raises heart rate and self-reported stress [30], so it provides a *non-emotional* arousal condition to contrast with the film-induced emotions.

3.9 N-back working-memory task (step 9, the other half)

Participant does	Watches a square appear in one of nine grid positions and presses *Match* when the position is the same as N trials earlier.
Parameters	8 practice trials with feedback, then 20 scored trials with exactly 30% targets and no accidental matches; stimulus 800 ms, onset-to-onset 1100 ms. The level N locks when practice starts. Hits, misses, false alarms, response time, and d' with the log-linear correction.
Measures	Working-memory load; physiological cost of sustained cognitive effort.
Why it is here	The N-back is the standard parametric working-memory task [21], with a well-mapped neural signature [29]; it is a valid, though not interchangeable, working-memory measure [18]. Performance is scored with signal detection theory [38]; the log-linear correction keeps d' finite for perfect scores and is the least biased of the standard corrections [15].

3.10 NASA-TLX workload rating (step 10)

Participant does	Rates mental, physical, and temporal demand, performance, effort, and frustration.
Parameters	Raw (unweighted) TLX, six 21-point scales (0–100 in steps of 5). About 45 s. Given after the cognitive task.
Measures	Perceived workload of the cognitive task.
Why it is here	The NASA Task Load Index is the most widely used subjective workload measure [14]; it checks that the cognitive tasks imposed the intended load, and lets physiological load be compared with perceived load.

3.11 Word list: study and recognise (step 23)

Participant does	Studies ten words for 20 s beside a countdown of the seconds left, then sees 20 words one at a time (the ten studied words mixed with ten look-alikes, e.g. *ocean*/*sea*) and answers Yes or No for each.
Parameters	Six lists of ten concrete, emotionally neutral nouns (one per semantic category, 4–8 letters, no colour words, no word shared between lists). 300 ms blank between test words; runs of at most 3. Scored as hits, false alarms, d', and response bias c.
Measures	Recognition memory after the emotional and cognitive blocks.
Why it is here	A yes/no recognition test scored with signal detection theory separates memory sensitivity from the tendency to say "yes" [36, 38], which typed free recall cannot do. Look-alike foils make the test sensitive to the difference between remembering the item and remembering its meaning. **[TODO: The lists were matched by hand; check them against published frequency and concreteness norms [8] before publication.]**

3.12 Read aloud (step 24)

Participant does Reads one fixed passage aloud, at a normal speaking pace, then presses *Done*.

Parameters 147 words on a single static screen, identical for every participant; nothing scrolls. The subject ends the step, and the reading time is recorded; it stops by itself after 120 s. Typically 50–70 s.

Measures Read speech: voice, articulation rate, and the breathing pattern imposed by reading, with the same text across the whole cohort.

Why it is here This is the protocol's speech condition. Speech is a hard case for contactless sensing, because it moves the face and imposes its own breathing pattern, so the dataset needs it. A fixed passage makes it comparable: every recording contains the same words in the same order, so voice and speech-rate measures can be compared across people, and so can the effect of speech on the contactless signals. The length is set from the reading-rate literature: the meta-analysis of 77 studies and 5,965 participants in [9] puts average *oral* reading at 183 words per minute, and reports lower rates for readers with English as a second language, so 147 words is about a minute across the range expected here. The text is deliberately neutral and concrete, because it sits inside an emotion protocol, and it shares no word with the Stroop colours or with the word-list nouns and their look-alikes, so it cannot prime either task. A unit test enforces both properties.

3.13 Controlled breathing (steps 28–30)

Participant does Breathes with a visual pacer at 6 and then at 20 breaths per minute, then follows a box-breathing pacer for 64 s (four rounds). All breathing screens use the same rising and falling marker and nothing else: no background video and no audio. In box breathing the marker climbs for four seconds, waits at the top for four, falls for four, and waits at the bottom, so the two holds are visible as pauses.

Parameters Each breathing condition shows its instructions first and starts only when the participant presses *Start* (the operator can also start it); its timer and pacer run from that press, which the export records as the step's onset. 6/min for 60 s (6 full cycles); 20/min for 42 s (14 cycles). Each paced rate enforces at least 6 cycles or 40 s, whichever is longer, so every condition is long enough to resolve a breathing rate spectrally. Box breathing runs 4 rounds, below that floor: it is a comfort and relaxation condition, not a step a breathing-rate estimator should be scored against.

Measures Breathing rate under known, controlled rates; heart-rate variability at slow breathing.

Why it is here These steps are the *ground truth* for estimating breathing rate from video, thermal imaging, and Wi-Fi CSI: the true rate is set by the pacer. The two rates bracket the normal range, from slow breathing at 6/min (near the resonance frequency used in HRV biofeedback, where heart-rate oscillations are largest [25]) to fast breathing at 20/min. Structured breathing practices such as box breathing measurably change physiological arousal [2]. They run after the two-minute post-exercise recovery (Section 2).

3.14 Stand-to-sit (step 25)

Participant does	With arms crossed on the chest, stands up fully and sits all the way back down, as many times as possible in one minute.
Parameters	60 s, timer-advanced, self-paced. The operator counts complete repetitions and adds the count as a session note. Skipped automatically if safety screening excludes exertion.
Measures	A rapid, graded rise in heart rate, breathing, and skin conductance, with large, repeated whole-body movement.
Why it is here	The one-minute sit-to-stand test is a standard, equipment-free test of functional exercise capacity; it produces a reproducible cardiorespiratory load within one minute [5]. Here its purpose is the load itself: it moves heart rate well above anything the seated tasks produce, which gives contactless heart-rate estimators a wider range to be validated over, and it adds a realistic motion condition. Being self-paced ("as many as you can"), the load differs between participants; the repetition count records by how much.

3.15 Step-up (step 26)

Participant does	Steps up onto a low step with one foot and then the other, then down the same way, at the fastest pace that feels comfortable.
Parameters	120 s, timer-advanced, self-paced, on a stable step 20–30 cm high placed beside the chair. The operator counts steps and adds the count as a session note. Skipped automatically if safety screening excludes exertion.
Measures	The session's highest heart rate, reached from an already raised level; the start of the recovery curve.
Why it is here	Step tests are the classic field method for loading the cardiovascular system in a controlled, submaximal way, and are used to estimate aerobic fitness [3]. Following the stand-to-sit directly, two minutes of stepping raises heart rate further, so the two exercises form a graded, increasing load. Unlike the standardized fitness tests, it is not metronome-paced here, because the aim is a high heart rate for sensor validation rather than a fitness estimate; the step count lets load be compared between participants.

3.16 Post-exercise recovery (step 27)

Participant does	Sits back down, stays still, and breathes normally.
Parameters	120 s, timer-advanced; a blood-pressure reading labelled "After exertion".
Measures	Heart-rate recovery at 1 and 2 minutes; the decay of breathing rate and skin conductance after exertion.
Why it is here	The fall in heart rate in the first minutes after exercise reflects cardiac parasympathetic reactivation and is a widely used autonomic marker; it is conventionally read at one and two minutes [10, 39]. A still, seated recording of two minutes captures exactly that window. Recovery is far from complete at two minutes; the breathing block and the body scan that follow run during the later part of it.

3.17 Guided body scan and return to normal (steps 31–32)

Participant does	Sits with eyes closed and follows four and a half minutes of guided audio, then opens the eyes and rests.
Parameters	Body scan 270 s (final guided audio, 4:29.5, `bodyscan-final-loud.mp4`, raised by 16 dB to about −16 LUFS so it is clearly audible at normal system volume); return to normal 20 s with a final blood-pressure reading.
Measures	A deliberately relaxed, low-arousal state at the end of the session, and the late part of the post-exercise recovery.
Why it is here	The body scan is a core practice of mindfulness-based stress reduction [20]. It gives the dataset a low-arousal reference after the tasks, complementing the resting baseline recorded before them, extends the recovery from the exercise, and ends the session calmly.

4 What each modality contributes

Modality	What it captures in this protocol	Key references
ECG chest strap	Reference heart rate and heart-rate variability for every step	[41, 24]
Finger oximeter (PPG)	Pulse rate and SpO_2; a second contact pulse reference	[1]
Skin conductance (EDA)	Sympathetic arousal: tonic level and phasic responses	[6]
Surface EMG	Muscle activity at 1000 Hz: facial or limb tension, and movement during the exercise steps	[16]
Face cameras (rPPG)	Contactless pulse from subtle skin-colour changes, plus facial expression. Camera 1: 640×480 at 30 fps, lossless, exposure fixed at 10 ms (one flicker cycle of 50 Hz mains lighting). Camera 2: 1080p at 25 fps, lossless, exposure fixed on the camera body	[42, 28]
Thermal camera	Facial temperature changes linked to arousal; breathing at the nostrils	[17]
Wi-Fi CSI (ESP32)	Contactless breathing and heart rhythm from radio channel changes	[26, 43, 45]
Audio (two mics)	Participant and operator speech in separate channels	[34]

Position among existing datasets. DEAP [22] and MAHNOB-HCI [37] pair film or music-video stimuli with contact physiology; WESAD [33] targets stress with wearables; UBFC-rPPG [4] provides face video with a pulse reference but no emotion. Meridian combines emotion induction, cognitive load, controlled breathing, and exercise-raised heart rate in one session, recorded simultaneously by contact and contactless sensors, including Wi-Fi CSI, on a common clock.

5 How version 26 differs from the full protocol

Compared with the full 44-step version, the shortened protocol removes:

- the eyes-closed resting baseline (90 s);
- the Part 2 and Part 3 cognitive tasks and their NASA-TLX ratings, so each participant does one task;

- the conversation task, which no participant now does;
- paced breathing at 10 and 15 breaths per minute (6 and 20 remain), and the rests between breathing conditions;
- the in-session personality questionnaire (TIPI now runs at consent) and the seated baseline before exercise;
- breath hold, head movement and speaking, standing, and the old 60 s brief exertion with its recovery.

It adds, since version 13: the read-aloud passage; the stand-to-sit (60 s) and step-up (120 s) exercises, which replace the old brief exertion with a longer, graded load; a two-minute post-exercise recovery; and a longer guided body scan (270 s, final audio). Paced breathing at 20/min runs 42 s. Return to normal is 20 s. Each return-to-baseline gate needs at least 25 s (the gate still needs physiology within tolerance for a continuous 15 s). Box breathing runs 64 s, four complete 16 s rounds, so it never ends mid-breath. The neutral control block (a neutral clip, its rating, and a washout after the resting baseline) has been removed from the template, because no neutral clip is in the stimulus library.

6 Open items for discussion

To resolve before data collection is final

1. **Fixed clip order.** All participants see the clips in the same order, so any change over the session (fatigue, habituation, drift) is confounded with the emotion shown. Record it as a limitation, or keep a within-part swap (e.g. disgust before surprise for half the participants) to recover some balance.
2. **Breathing after exercise.** The paced-breathing steps start about two minutes after the step-up, during late recovery, so heart rate, breathing drive, and HRV are not yet back at rest [39]. They remain valid as a breathing-*rate* reference if the participant can follow the pacer, but slow-breathing HRV measured there is not a resting value. Check pacer adherence at 6/min in the first sessions.
3. **Planning estimates.** The builder sets the word list at 30 s and the read-aloud at 30 s. The word list alone needs 20 s of study plus 20 timed test words, and the 147-word passage typically takes 50–70 s to read, so real sessions will likely run about a minute longer than the 37 min shown.
4. **One cognitive task per participant.** Stroop and N-back are now each done by half of the cohort, so task effects can only be compared between participants, and each task has half the recordings it had when everyone did both. Confirm this is intended.
5. **Film set.** Name and cite the validated set the six clips come from: **[TODO: film set]**.
6. **Word lists.** Check the six lists against published frequency and concreteness norms [8].
7. **Safety screening.** Screening matters again: a yes to the heart, breathing, mobility, pregnancy, or "unwell today" questions skips both exercises, and the skip is logged. The fainting question currently skips nothing, because no step is tagged as a standing step; decide whether it should also skip the stand-to-sit.
8. **Washout length and a gate after Part 3.** The 20 s washout between the two clips of a part is shorter than most film-elicitation protocols use (about one to two minutes) [31], and Part 3 has no return-to-baseline gate before the word list and read-aloud. Lengthening the washouts to 60 s would add 2:40; a Part 3 gate about 30 s.

9. **Rig status at the sensor check of 21 September 2026.** Skin conductance now gives a live signal (0–12.9 μS), but still needs a known-resistor check. Surface EMG is dominated by 50 Hz mains pickup (85–97% of in-band power) through the mains-earthed host; the fix is galvanic isolation, not filtering. In the three-minute test the oximeter, the ECG stream, and the CSI receiver delivered no data and must be checked before each session. There is still no respiration belt, so the paced-breathing steps have no independent breathing reference, and no photodiode to measure display latency. The second face camera's exposure cannot be locked over USB and must be set to manual on the camera body.

References

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